As ΔA /A, ΔB / B are both small quantities, their product term $\frac{\Delta A}{A}.\frac{\Delta B}{B}$ can be neglected. The maximum fractional error in Z is

$$\frac{\Delta Z}{Z} = \pm\left(\frac{\Delta A}{A} + \frac{\Delta B}{B}\right) \qquad (1.5)$$

The maximum fractional error in the product of two quantities is equal to the sum of the fractional errors in the individual quantities.

[Alternative method is given in Appendix A1.2]

EXAMPLE 1.7

The length and breadth of a rectangle are (5.7 ± 0.1) cm and (3.4 ± 0.2) cm respectively. Calculate the area of the rectangle with error limits.

Solutions

Length $\ell = (5.7 \pm 0.1)$ cm

Breadth b =(3.4 ± 0.2) cm

Area A with error limit = A ± ΔA = ?

Area A = $\ell \times b = 5.7 \times 3.4 = 19.38 = 19.4\ cm^2$

$$\frac{\Delta A}{A} = \frac{\Delta l}{l} + \frac{\Delta b}{b}$$

$$\Delta A = \left(\frac{\Delta l}{l} + \frac{\Delta b}{b}\right)A$$

$$\Delta A = \left(\frac{0.1}{5.7} + \frac{0.2}{3.4}\right)19.4$$

$$= (0.0175+0.0588) \times 19.4$$

$$= 1.48 = 1.5$$

Area with error limit

A = (19.4±1.5) cm^2

(iv) Error in the division or quotient of two quantities

Let ΔA and ΔB be the absolute errors in the two quantities A and B respectively.

Consider the quotient, $Z = \frac{A}{B}$

The error ΔZ in Z is given by

$$Z \pm \Delta Z = \frac{A \pm \Delta A}{B \pm \Delta B} = \frac{A\left(1 \pm \frac{\Delta A}{A}\right)}{B\left(1 \pm \frac{\Delta B}{B}\right)}$$

$$= \frac{A}{B}\left(1 \pm \frac{\Delta A}{A}\right)\left(1 \pm \frac{\Delta B}{B}\right)^{-1}$$

or $Z \pm \Delta Z = Z\left(1 \pm \frac{\Delta A}{A}\right)\left(1 \mp \frac{\Delta B}{B}\right)$ [using $(1+x)^n \approx 1+nx$, when $x<<1$]

Dividing both sides by Z, we get,

$$1 \pm \frac{\Delta Z}{Z} = \left(1 \pm \frac{\Delta A}{A}\right)\left(1 \mp \frac{\Delta B}{B}\right)$$

$$= 1 \pm \frac{\Delta A}{A} \mp \frac{\Delta B}{B} \pm \frac{\Delta A}{A}.\frac{\Delta B}{B}$$

As the terms ΔA/A and ΔB/B are small, their product term can be neglected.

The maximum fractional error in Z is given by $\frac{\Delta Z}{Z} = \left(\frac{\Delta A}{A} + \frac{\Delta B}{B}\right)$ (1.6)

The maximum fractional error in the quotient of two quantities is equal to the sum of their individual fractional errors.

(Alternative method is given in Appendix A1.2)

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